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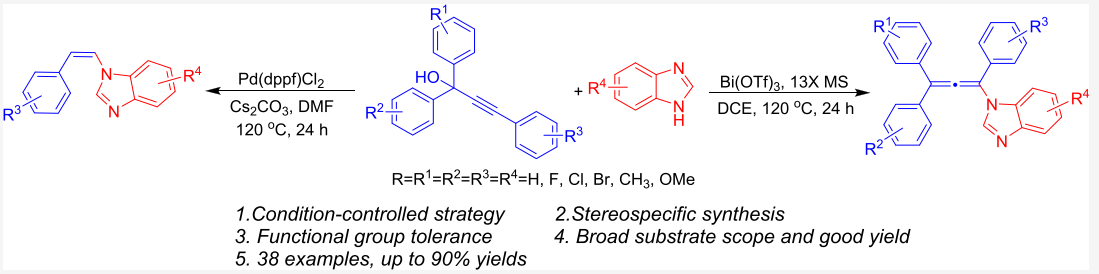
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Condition-Controlled Divergent Selective Synthesis of (*Z*)-*N*-Vinyl and *N*-Allenyl Benzimidazoles by Pd- or Bi-Catalyzed Direct *N*-Alkenylation Reactions

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Cite This: *J. Org. Chem.* 2025, 90, 2602−2612 Read Online

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ABSTRACT: We have developed a condition-controlled divergent synthesis of (*Z*)-*N*-vinyl and *N*-allenyl benzimidazoles from 1,1,3-triphenylprop-2-yn-1-ols and benzimidazoles through Pd- or Bi-catalyzed *N*-alkenylation reactions involving nucleophilic attack and C−C bond cleavage processes. The desired two different kinds of products can be conveniently and selectively synthesized by using this strategy, which features stereospecific synthesis, good functional group tolerance, a broad substrate scope, and high efficiency. The strategy provides significant advantages for the synthesis of biologically and pharmaceutically active imidazoheterocycles.

INTRODUCTION

Benzimidazoles[1] and allenes[2] are two important classes of organic frameworks and are widely found in natural compounds, pharmaceuticals, organic intermediates, and materials science due to their unique biological activities and physical properties. For example, as shown in Figure 1, *N*-vinyl benzimidazoles (**A**) and (**B**) exhibit potent PPARc partial agonistic activity and modest PPARa agonistic activity.[3] *R*-Cytallene (**C**) and *R*-adenallene (**D**) are inhibitors of HIV and hepatitis B replication.[4] The natural product peridinin (**E**) is the main light-absorbing pigment in peridinin-chlorophyll *α*-protein (PCP), and its unique structure significantly alters the character of excited states.[5] However, new types of functionalized *N*-allenyl benzimidazoles, which possess potential research values, have very few relevant reports.[6] Therefore, the synthesis of *N*-vinyl and *N*-allenyl benzimidazoles holds certain research significance.

Metal-catalyzed *N*-alkenylation is a prominent method for C−N bond formation,[7] serving as an efficient strategy for the direct and precise synthesis of organic nitrogen-containing intermediates.[8] Over the past decades, numerous metal catalytic systems have been explored.[9] Recently, Kikuchi and Yoshikai and co-workers reported *N*-alkenylation reactions of azoles, alkynes, and iodine(III) electrophiles for the synthesis of *N*-vinylazoles.[10] Anderson and co-workers developed a gold-catalyzed *N*-alkenylation of isoxazolines to synthesize *N*-vinyl gold intermediates for further derivatization.[11] The direct *N*-alkenylation of benzimidazoles has been achieved and obtained good results,[12] but there are relatively few reports on *N*-alkylation catalyzed by bismuth metal. Herein, seeking new catalytic reactions to assemble various functionalized (*Z*)-*N*-vinylbenzimidazoles with economically accessible starting materials is still very meaningful (Scheme 1).[13] Considering the significant advances in the functionalization of allenes and benzimidazoles, it would be meaningful for the convenient assembly of new *N*-allenyl benzimidazoles.

The metal-catalyzed nucleophilic attack and C−C bond cleavage processes can efficiently facilitate the *N*-alkenylation reaction.[14] By involving metal catalysis, nucleophilic attack reactions proceed mildly and efficiently, leading to the formation of C−N bonds.[15] Additionally, metal-catalyzed C−C bond cleavage has emerged as an exceptionally efficient and powerful method for breaking inert and thermodynamically stable C−C bonds.[16] This reaction can participate in numerous subsequent transformations and maintain high regioselectivity

Received: October 10, 2024
Revised: January 20, 2025
Accepted: January 29, 2025
Published: February 6, 2025



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https://doi.org/10.1021/acs.joc.4c02516
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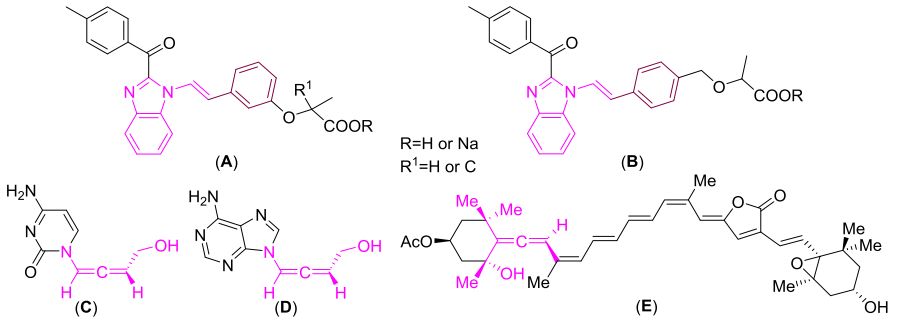
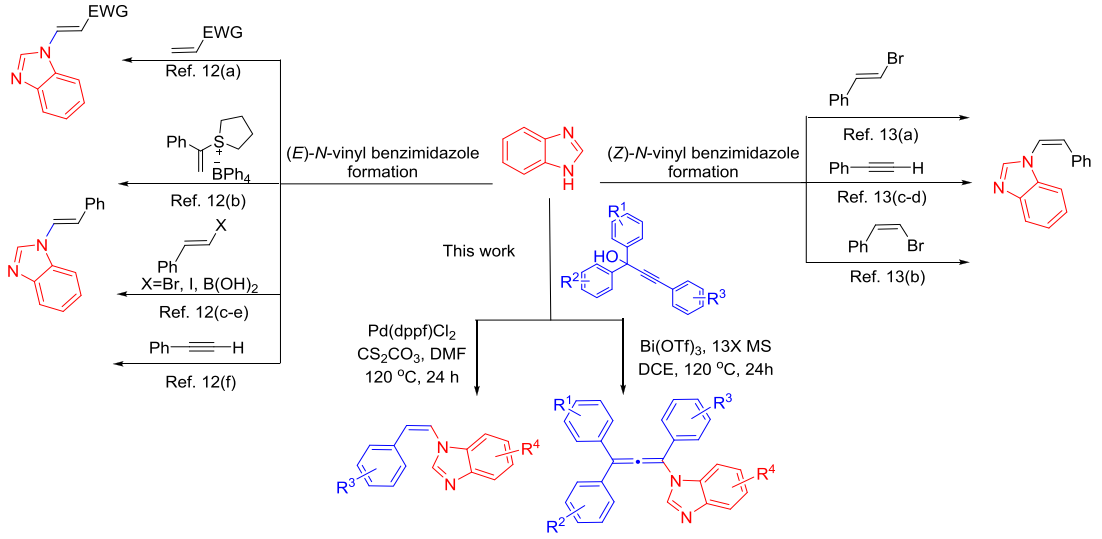


Figure 1. Some useful (*Z*)-*N*-vinyl benzimidazoles and allenes.

Scheme 1. Background of This Work



and stereospecificity.[17] Furthermore, propargylic alcohol chemistry is a powerful tool for constructing various complex structural frameworks for the bifunctional properties of propargylic alcohols.[18] Herein, as a continuation of our previous works, we reported a condition-controlled strategy for the selective synthesis of (*Z*)-*N*-vinyl and *N*-allenyl benzimidazoles from 1,1,3-triphenylprop-2-yn-1-ols and benzimidazoles through Pd- or Bi-catalyzed *N*-alkenylation reactions via nucleophilic attack and C−C bond cleavage processes.

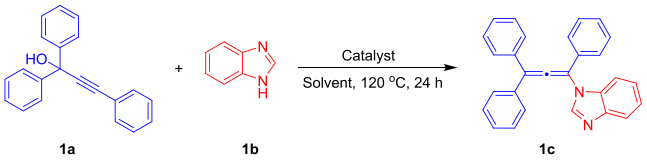
RESULTS AND DISCUSSION

Initially, we searched for suitable conditions for the synthesis of 1-(1,3,3-triphenylpropa-1,2-dien-1-yl)-1*H*-benzo[*d*]-imidazole **1c** using 1,1,3-triphenyl-2-acetyl-1-ol **1a** and benzimidazole **1b** as the model substrates (Table 1). The reaction was carried out with $Yb(OTf)_3$ as the catalyst in DCE at 120 °C for 24 h. The anticipated product **1c** was isolated in a 33% yield (Table 1, entry 1). We screened different Lewis acids, and $Bi(OTf)_3$ was the best choice (Table 1, entries 1−6). Next, different solvents were tested (DMSO, $CHCl_3$, MeCN, DMF, and DCE), and the use of DCE afforded the highest yield (Table 1, entries 5 and 7−10). When 1.2 equiv of **1a** was used, the yield increased significantly (Table 1, entry 11). To our delight, the product was obtained in good yield when 13X molecular sieve (80 mg) was used, indicating that anhydrous conditions are important for the reaction (Table 1, entry 12). At the same time, we also carried out the gram-level experiment, and the reaction yield did not change (80%, 0.61g; Table 1, entry 12). We used $Cu(OTf)_2$ as a catalyst to replace $Bi(OTf)_3$ under the best conditions (Table 1, entry 13), and **1c** was isolated in 42%. Without the catalyst, no product was detected (Table 1, entry 14).

With the optimized reaction conditions in hand, some representative substrates were selected to explore the scope of this transformation (Table 2). First, we evaluated the influence of substituted benzimidazoles **b**. The yields decreased when substrates bearing electron-withdrawing functional groups were employed (Table 2, **1c−5c**). Interestingly, when 7-methyl-1*H*-benzo[*d*]imidazole was used to do the reaction, only 4-methyl-1-(1,3,3-triphenylpropa-1,2-dien-1-yl)-1*H*-benzo[*d*]imidazole **2c** was isolated for the steric effect. The HMBC confirmed the structure of **2c** (for, see ESI, Figures S3 and S4). The reactions of 2-methyl and 2-phenyl benzimidazoles proceeded smoothly, and the corresponding products were isolated in moderate to good yields (Table 2, **6c−7c**). The electronic effect of R^3 greatly influenced the transformation (Table 2, **8c−10c**). When R^3 was an electron-withdrawing functional group (-Cl), no product was detected (Table 2, **11c**). Other substituted substrates **a** with different functional groups (R^2, R^3 = CH_3, Cl, OCH_3) were suitable for the reaction, and the products **c** were also isolated in moderate to good yields (Table 2, **12c−21c**). When R^4 was a weak electron-donating functional group (5-CH_3), the mixtures of regioisomers were obtained (Table 2, **3c** + **3c′** and **15c** + **15c′**). On the other hand, when R^4 was an electron-withdrawing functional group (-Cl), only the product **4c** was isolated for the electronic effect (Table 2, **4c**). However, the

https://doi.org/10.1021/acs.joc.4c02516
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Table 1. Optimization of the Reaction Conditions for the Synthesis of 1c[a,b]



1a 1b 1c

entry	catalyst	solvent	yield (%)
1	$Yb(OTf)_3$	DCE	33
2	$Al(OTf)_3$	DCE	50
3	$Zn(OTf)_3$	DCE	46
4	$Sc(OTf)_3$	DCE	40
5	$Bi(OTf)_3$	DCE	58
6	AgOTf	DCE	50
7	$Bi(OTf)_3$	DMSO	trace
8	$Bi(OTf)_3$	$CHCl_3$	trace
9	$Bi(OTf)_3$	MeCN	trace
10	$Bi(OTf)_3$	DMF	trace
11	$Bi(OTf)_3$	DCE	65[c]
12	$Bi(OTf)_3$	DCE	80[d,e]
13	$Cu(OTf)_2$	DCE	42[d]
14		DCE	0

[a]Reaction conditions: catalyst (0.045 mmol, 0.15 equiv), **a** (0.3 mmol, 1.0 equiv), **b** (0.3 mmol, 1.0 equiv) in solvent (2.0 mL) at 120 °C for 24 h. [b]The yields were determined by ^{1}H NMR analysis by using 1,3,5-trimethoxybenzene as the internal standard. [c]**1a** (0.36 mmol, 1.2 equiv) was used. [d]**1a** (0.36 mmol, 1.2 equiv) and 13× molecular sieve (80 mg) were added. [e]**1a** (2.4 mmol, 1.2 equiv), **1b** (2 mmol, 1.0 equiv), and 13× molecular sieve (0.53 g) were used.

reaction of imidazole failed to generate the corresponding product (Table 2, **22c**). The above results revealed that the reaction was mainly influenced by the electronic effect.

Next, we tried to optimize the reaction conditions for the synthesis of (*Z*)-1-styryl-1*H*-benzo[*d*]imidazole **1d** using **1a** and **1b** as the substrates (Table 3). We screened different solvents (DMF, *p*-xylene, and toluene), catalysts [$Pd(OAc)_2$/PPh_3, $PdCl_2/PPh_3$, $Pd(dppf)Cl_2$, $Pd(OAc)_2$, $Pd(PPh_3)_4$, and $Pd(dba)_2$], and reaction temperatures for the reaction (Table 3, entries 1−12). The best set of reaction conditions was using $Pd(dppf)Cl_2$ (0.15 equiv) as the catalyst, Cs_2CO_3 (2.0 equiv) as the base in DMF at 120 °C for 24 h. X-ray crystallography confirmed the structures of **1d** and **4d** (CCDC 2364113 and 2364114). After the optimal conditions were established, the substrate range for the synthesis of **d** was examined (Table 4). The yields were much higher when **b** bearing electron-donating functional groups were utilized (Table 4, **1d**−**5d**). The HMBC confirmed the structure of **2d** (for, see ESI, Figures S5 and S6), indicating that the reactions were influenced by the steric effect of **b.** The structure of **d** was determined by the functional groups R^3 and R^4. When R^1 and R^2 were assigned different functional groups, the yields did not vary obviously. R^3 with electron-donating functional groups produced much higher yields (Table 4, **6d**−**10d**). Unfortunately, imidazole was not suitable for the reaction, and no corresponding products were isolated (Table 4, **11d**). From these results, it could be found that the reaction was also influenced by the electronic effect, and R^3 and R^4 with electron-donating functional groups were more favorable for the reaction. For the Pd-catalyzed *N*-alkenylation reactions via nucleophilic addition and C−C bond cleavage processes, no (*E*)-products were observed during the synthesis, indicating the advantages of this method for efficient stereospecific synthesis.

We conducted several control experiments that could support the mechanisms of the reactions (Scheme 2). First, **1a** was transformed to benzophenone **1h** and 1,4-diphenylbuta-1,3-diyne **1i** under the optimal reaction conditions for the synthesis of **1d** (eq 1). Then, phenylacetylene **1e** and **1b** were also reacted under the optimal conditions, and only (*E*)-1-styryl-1*H*-benzo[*d*]imidazole **1f** was isolated in 70% yield (eq 2), which confirmed the Pd-catalyzed nucleophilic addition and C−C bond cleavage processes. Finally, deuterated DMF was used as the reaction solvent, and **1d** was detected without deuterated H in 75% yield (eq 3), excluding the proton exchange from DMF during the reaction.

Based on experimental observation and related literature,[19] the mechanisms are proposed in Scheme 3. For *N*-allenyl benzimidazoles, intermediate **1aa** was formed by the coordination of $Bi(OTf)_3$ with **1a**. Then, **1b** could react with **1aa** to form intermediate **1ab** by nucleophilic attack reaction, and **1c** was produced from intermediate **1ab** by the elimination of $Bi(OTf)_3$ and H_2O. For the synthesis of (*Z*)-*N*-vinyl benzimidazoles, organometallic intermediate **1ac** could be formed first; **1ad** was obtained by the nucleophilic addition reaction of **1ac** with **1bb**, which was formed via **1b** and Base. After the transmetalation and protonolysis processes, **1ad** was transferred to **1ae** and **1af**. Intermediate **1ag** was produced by the C−C bond cleavage process, and **1d** was obtained in the present of $BaseH^+$. When comparing the two reactions, the formation of the five-membered ring intermediate **1ad** of the palladium-catalyzed reaction was stable, so **1bb** attacked the alkynyl carbon near the hydroxyl group. The Bi(III) catalyst probably only plays a coordination role and promotes hydroxyl departure. The electronic effect makes the phenyl-linked alkynyl carbon more electrophilic, favoring nucleophilic attack.

CONCLUSIONS

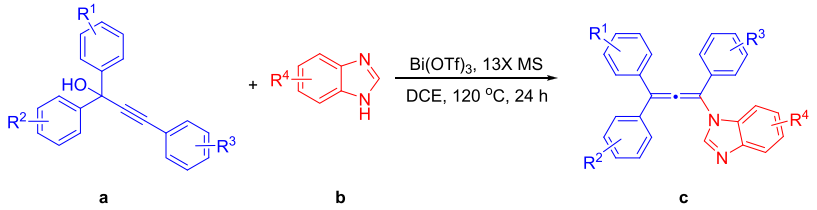
In summary, we have developed Pd- or Bi-catalyzed *N*-alkenylation reactions for the selective synthesis of (*Z*)-*N*-vinyl and *N*-allenyl benzimidazoles from 1,1,3-triphenylprop-2-yn-1-ols and benzimidazoles via nucleophilic attack and C−C bond cleavage processes. The reactions exhibit good stereoselectivity, broad functional group tolerance, and good yields, and a wide variety of (*Z*)-*N*-vinyl and *N*-allenyl benzimidazoles were obtained. The importance of these substituted benzimidazoles would make the approach very attractive for biological and pharmaceutical applications.

EXPERIMENTAL SECTION

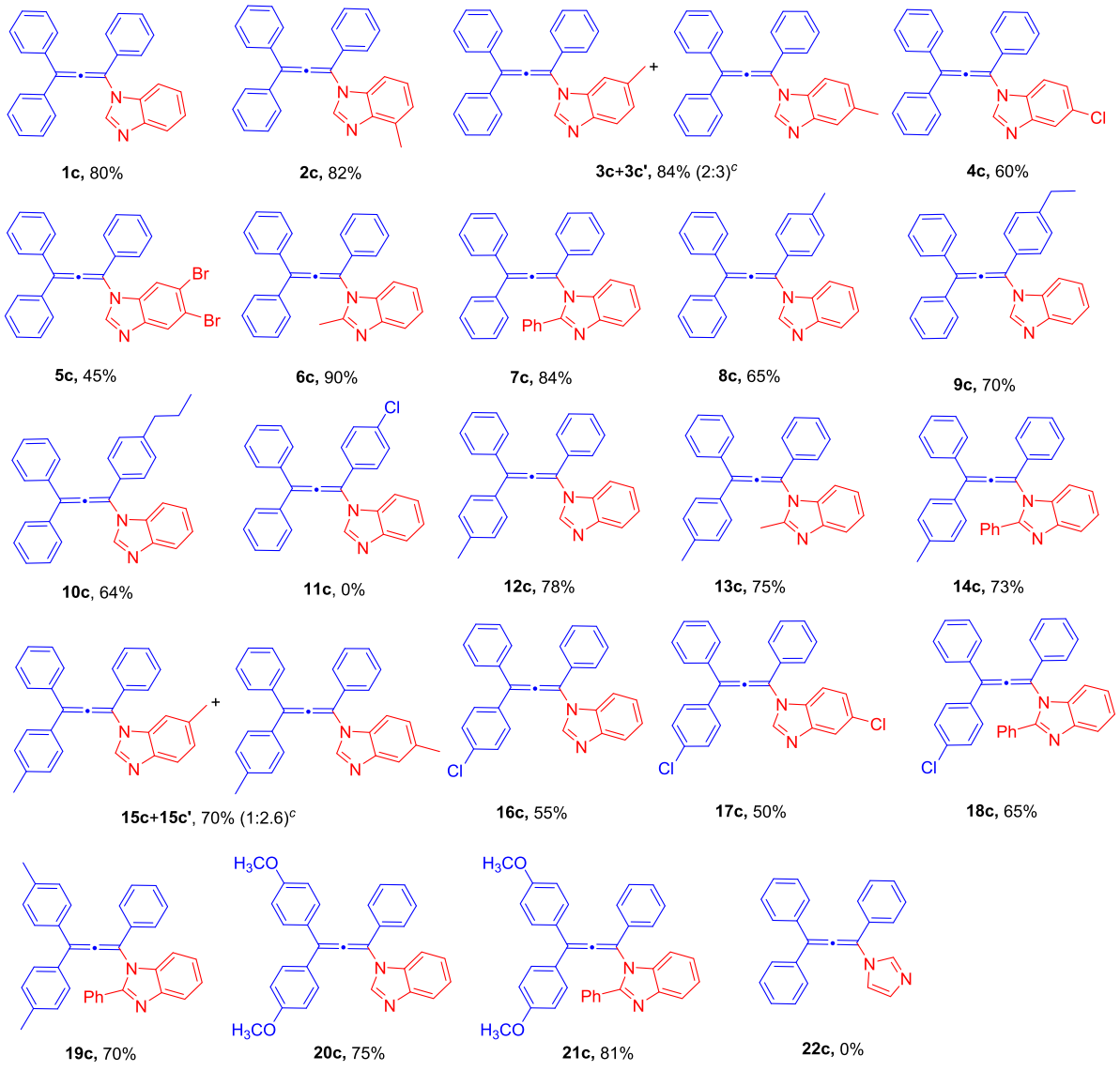
General Remarks. ^{1}H NMR and $^{13}C\{^1H\}$ NMR spectra were recorded on a 500 MHz (Bruker) instrument, and $CDCl_3$ (δ_H = 7.26 ppm, δ_C = 77.16 ppm) was used as the internal standard. Chemical shifts were reported in ppm. Multiplicity was recorded: s (singlet), d (doublet), t (triplet), q (quartet), dd (doublet of doublets), dt (doublet of triplets), and m (multiplet). The direct used reagents and solvents were of pure analytical grade and purchased from commercial sources, if not stated otherwise. The starting substrates were synthesized according to the known literature.[12h] Column chromatography was hand-packed with silica gel (200−300 mesh). Melting points were uncorrected. High-resolution mass spectra (HRMS) were recorded on a Q-TOF Premier instrument (ESI). Silica gel plates (GF254, 0.2 mm thick) were used for TLC testing. An oil bath was used as the heat source for reactions that require heating. X-ray diffraction patterns were collected by a Rigaku D/max-2550 diffractometer with CuKα radiation.

https://doi.org/10.1021/acs.joc.4c02516
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Table 2. Substrate Scope of the *N*-Alkenylation for the Synthesis of *N*-Allenyl Benzimidazoles[a,b]



a b c



[a]Reaction conditions: catalyst (0.045 mmol, 0.15 equiv), **a** (0.36 mmol, 1.2 equiv), **b** (0.3 mmol, 1.0 equiv), and 13× molecular sieve (80 mg) in DCE (2.0 mL) reacted at 120 °C for 24 h. [b]Isolated yields after flash chromatography based on **b**. [c]A mixture of two isomers was obtained (the isomeric ratio was approximately determined by ^{1}H NMR).

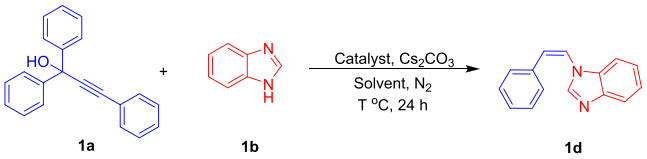
Safety Statement. Caution! *n*-BuLi solution is a highly flammable liquid. It can spontaneously ignite upon exposure to air and release flammable gases upon contact with water. It must be handled with great care.

General Procedure for the Synthesis of 1,1,3-Triphenylprop-2-yn-1-ols.[20] Phenylacetylenes (2.04g, 20 mmol) were dissolved in anhydrous tetrahydrofuran (100 mL) and stirred at −78 °C for 10 min. Then, *n*-BuLi (15 mL, 24 mmol) was added dropwise, and the mixture was stirred for 0.5 h. Next, diphenylmethanones (3.64 g, 20 mmol) were added and stirred for 1 h. Finally, the reaction was allowed to continue at room temperature for 4 h. After the reaction was completed, saturated ammonium chloride solution was added to quench the reaction. The mixture was extracted with ethyl acetate three times. The organic phase was dried over Na_2SO_4, filtered, and concentrated under reduced pressure. The crude oil mixture was purified by silica gel column chromatography (petroleum PE/EtOAc = 4:1) to obtain 1,1, 3-triphenylpropane-2-acetylene-1-ols.

General Procedure for the Synthesis of c (1c−22c). The reaction was conducted with 1,1,3-triphenylprop-2-yn-1-ols **a** (0.36 mmol, 1.2 equiv), 1*H*-benzo[*d*]imidazoles **b** (0.3 mmol, 1.0 equiv), $Bi(OTf)_3$ (0.045 mmol, 0.15 equiv), and DCE (2.0 mL) in a Schlenk

https://doi.org/10.1021/acs.joc.4c02516
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Table 3. Optimization of the Reaction Conditions for the Synthesis of 1d[a,b]



1a 1b 1d

entry	catalyst	solvent	T (°C)	yield (%)
1	$Pd(OAc)_2/PPh_3$	DMF	120	53
2	$Pd(OAc)_2/PPh_3$	DMF	135	53
3	$Pd(OAc)_2/PPh_3$	DMF	110	43
4	$Pd(OAc)_2/PPh_3$	*p*-xylene	120	0
5	$Pd(OAc)_2/PPh_3$	DMSO	120	43
6	$Pd(OAc)_2/PPh_3$	toluene	120	0
7	$PdCl_2/PPh_3$	DMF	120	16
8	$Pd(dppf)Cl_2$	DMF	120	75[c]
9	$Pd(OAc)_2$	DMF	120	13
10	$Pd(PPh_3)_4$	DMF	120	0
11	$Pd(dba)_2$	DMF	120	0
12		DMF	120	0

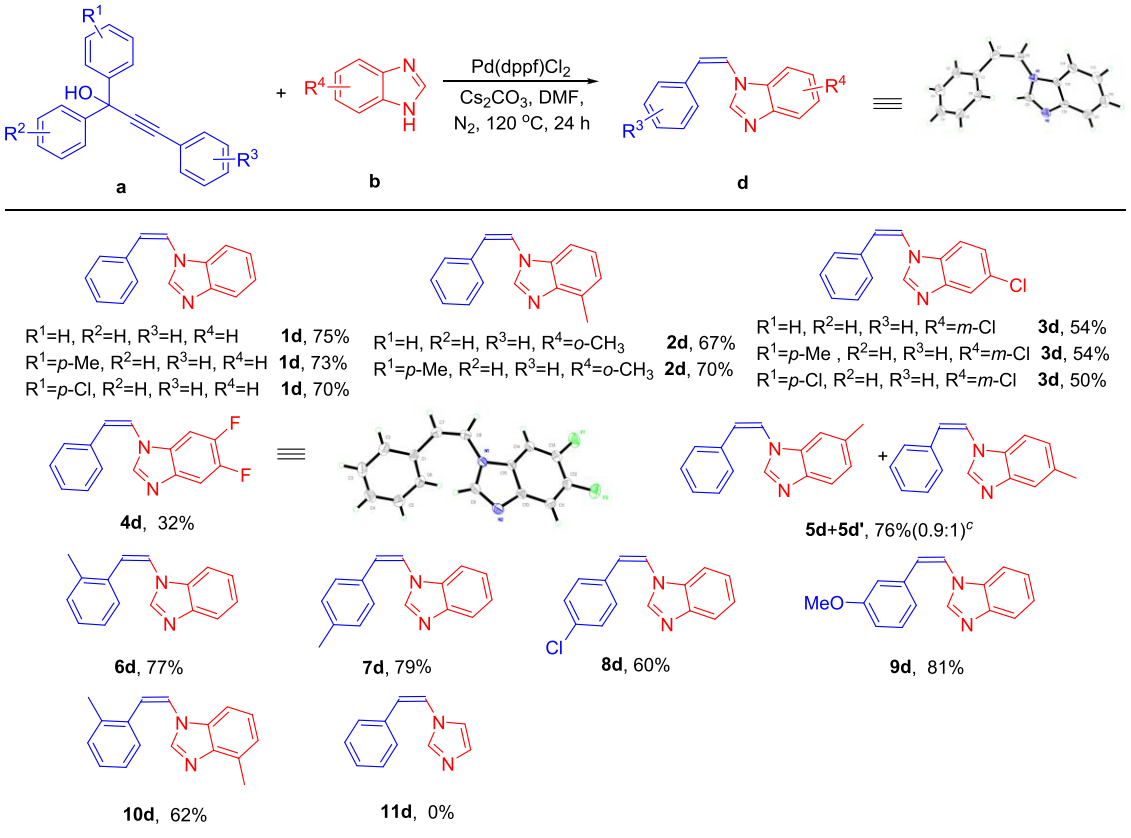
[a]Reaction conditions: catalyst (0.045 mmol, 0.15 equiv), **a** (0.3 mmol, 1.0 equiv), **b** (0.3 mmol, 1.0 equiv), and Cs_2CO_3 (0.6 mmol, 2.0 equiv) in solvent (2.0 mL) under N_2 for 24 h. [b]Isolated yields after flash chromatography based on **b**. [c]Diphenylmethanone **1h** was isolated in 80% yield.

tube at 120 °C for 24 h. After the reaction was completed, the mixture was cooled to room temperature. The reaction solution was filtered and washed with EtOAc (30 mL) three times. The organic phase was concentrated, and the desired products **c** were purified by silica gel column chromatography using PE/EtOAc as the eluent.

General Procedure for the Synthesis of d (1d−11d). The reaction was conducted with 1,1,3-triphenylprop-2-yn-1-ols **a** (0.3 mmol, 1.0 equiv), 1*H*-benzo[*d*]imidazoles **b** (0.3 mmol, 1.0 equiv), $Pd(dppf)Cl_2$ (0.045 mmol, 0.15 equiv), and Cs_2CO_3 (0.6 mmol, 2.0 equiv) in a Schlenk tube equipped with Teflon valves. The tube was placed under vacuum for 20 min, refilled with nitrogen, and then DMF (2.0 mL) was added. The reaction was carried out at 120 °C for 24 h. After the reaction was completed, it was cooled to room temperature. H_2O (30 mL) was added, the mixture was extracted with EtOAc (20 mL × 3), and the combined EtOAc was washed with brine (20 mL × 3). The organic phase was concentrated and purified using silica gel column chromatography to obtain product **d**. Single crystals of **1d** and **4d** were obtained by slow diffusion of petroleum ether into the dichloromethane solution of the compounds.

1-(1,3,3-Triphenylpropa-1,2-dien-1-yl)-1H-benzo[d]imidazole (1c). Following is the topical procedure for the synthesis of **1c**: **1a** (0.36 mmol), **1b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (92.2 mg, 80%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 8.06 (s, 1H), 7.88 (d, J = 8.5 Hz, 1H), 7.42−7.36 (m, 13H), 7.32−7.29 (m, 3H), 7.20−7.13 (m, 2H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$/TMS) δ: 204.1, 143.3, 135.2,

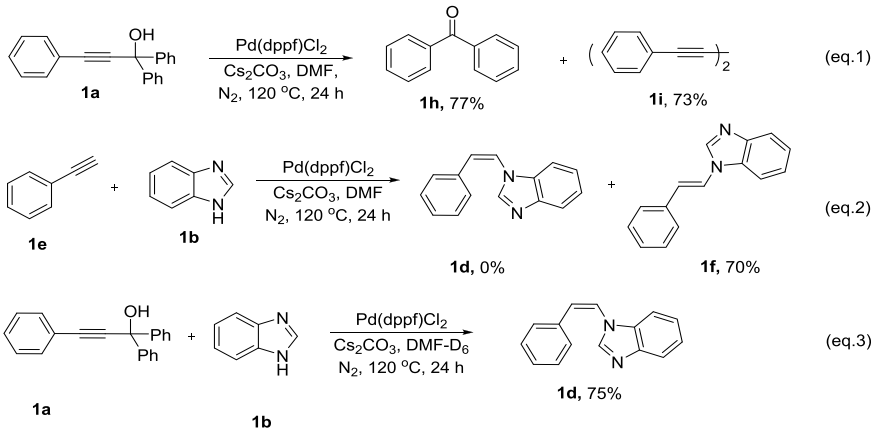
Table 4. Substrate Scope of the *N*-Alkenylation for the Synthesis of (*Z*)-*N*-Vinyl Benzimidazoles[a,b]



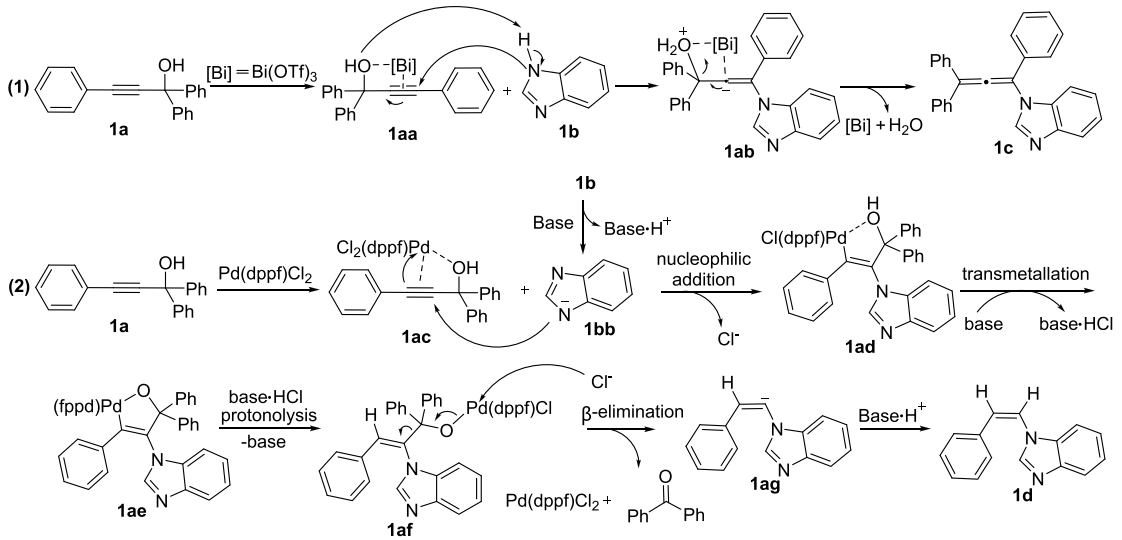
[a]Reaction conditions: $Pd(dppf)Cl_2$ (0.045 mmol, 0.15 equiv), **a** (0.3 mmol, 1.0 equiv), **b** (0.3 mmol, 1.0 equiv), and Cs_2CO_3 (0.6 mmol, 2.0 equiv) in DMF (2.0 mL) reacted at 120 °C under a N_2 atmosphere for 24 h. [b]Isolated yield after flash chromatography based on **b**. [c]A mixture of **5d** was obtained (the isomeric ratio was approximately determined by ^1H NMR).

https://doi.org/10.1021/acs.joc.4c02516
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Scheme 2. Reaction Mechanism Studies



Scheme 3. Proposed Mechanisms



132.8, 129.3, 129.1, 129.0, 128.9, 128.9, 128.6, 127.1, 126.3, 123.7, 122.8, 120.6, 119.4, 111.5, 111.1. HRMS (ESI) *m/z*: $[M + H]^+$ Calcd for $C_{28}H_{21}N_2$ 385.1699; Found 385.1692.

*4-Methyl-1-(1,3,3-triphenylpropa-1,2-dien-1-yl)-1H-benzo[d]-imidazole (**2c**).* Following is the topical procedure for the synthesis of **2c**: **1a** (0.36 mmol), (4-methyl-1*H*-benzo[*d*]imidazole), **2b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (95.6 mg, 82%) as an orange-red oily liquid. TLC: $R_f \approx 0.6$. ^{1}H NMR (500 MHz, $CDCl_3$/TMS) δ 8.03 (s, 1H), 7.41−7.35 (m, 13H), 7.31−7.29 (m, 2H), 7.11−7.10 (m, 1H), 7.09−7.06 (m, 1H) 6.97 (d, J = 7.5 Hz 1H), 2.73 (s, 3H). ^{13}C{^{1}H}·NMR (126 MHz, $CDCl_3$/TMS) δ: 204.2, 143.0, 142.4, 135.2, 134.0, 132.8, 130.5, 129.2, 129.1, 128.9, 128.9, 128.8, 126.3, 123.6, 123.2, 119.2, 111.2, 109.0, 16.8. HRMS (ESI) *m/z*: $[M + H]^+$ Calcd for $C_{29}H_{23}N_2$ 399.1856; Found 399.1849.

*6-Methyl-1-(1,3,3-triphenylpropa-1,2-dien-1-yl)-1H-benzo[d]-imidazole and 5-Methyl-1-(1,3,3-triphenylpropa-1,2-dien-1-yl)-1H-benzo[d]imidazole (**3c + 3c′**).* Following is the topical procedure for the synthesis of **3c** + **3c′**: **1a** (0.36 mmol), (5-methyl-1*H*-benzo[*d*]imidazole), **3b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (100.34 mg, 84%) as a dark yellow oily liquid. TLC: $R_f \approx 0.6$. ^{1}H NMR (500 MHz, $CDCl_3$/TMS) δ 7.99 (d, J = 5.5 Hz, 1H), 7.73 (d, J = 8.5 Hz, 0.5H), 7.65 (s, 0.4H), 7.41−7.36 (m, 13.5H), 7.32−7.29 (m, 2H), 7.11 (d, J = 8.0 Hz, 0.7H), 7.00 (s, 1H), 6.85 (s, 0.6H), 2.48 (s, 1.3H), 2.27 (s, 1.7H). ^{13}C{^{1}H}·NMR (126 MHz, $CDCl_3$/TMS) δ 204.3, 204.1, 144.1, 143.2, 142.7, 141.7, 135.2, 135.2, 134.5, 133.7, 132.9, 132.8, 132.6, 129.2, 129.2, 129.1, 128.9, 128.9, 128.9, 128.9, 126.3, 126.3, 125.2, 124.4, 120.3, 120.0, 119.3, 119.2, 111.4, 111.2, 111.1, 111.0, 21.8, 21.6. HRMS (ESI) *m/z*: $[M + H]^+$ Calcd for $C_{29}H_{23}N_2$ 399.1856; Found 399.1857.

*5-Chloro-1-(1,3,3-triphenylpropa-1,2-dien-1-yl)-1H-benzo[d]-imidazole (**4c**).* Following is the topical procedure for the synthesis of **4c**: **1a** (0.36 mmol), (5-chloro-1*H*-benzo[*d*]imidazole), **4b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (75.3 mg, 60%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^{1}H NMR (500 MHz, $CDCl_3$/TMS) δ 7.92 (s, 1H), 7.71 (d, J = 2.0 Hz, 1H), 7.28−7.23 (m, 13H), 7.15−7.13 (m, 2H), 7.00 (dd, J_1 = 8.5 Hz J_2 = 1.5 Hz, 1H), 6.88 (d, J = 8.5 Hz, 1H). ^{13}C{^{1}H}·NMR (126 MHz, $CDCl_3$/TMS) δ: 203.9, 144.6, 144.3, 134.9, 132.9, 132.4, 129.3, 129.3, 129.0, 129.0, 128.9, 128.5, 126.2, 124.2, 120.4, 119.7, 112.2, 110.8. HRMS (ESI) *m/z*: $[M + H]^+$ Calcd for $C_{28}H_{20}N_2Cl$ 419.1310; Found 419.1308.

*5,6-Dibromo-1-(1,3,3-triphenylpropa-1,2-dien-1-yl)-1H-benzo-[d]imidazole (**5c**).* Following is the topical procedure for the synthesis of **5c**: **1a** (0.36 mmol), (5,6-dibromo-1*H*-benzo[*d*]imidazole), **5b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (72.9 mg, 45%) as a light green oily liquid. TLC: $R_f \approx 0.6$. ^{1}H NMR (500 MHz, $CDCl_3$/TMS) δ: 8.13 (s, 1H), 8.03 (s, 1H), 7.44−7.37 (m, 15H), 7.28−7.27 (m, 1H). ^{13}C{^{1}H}·NMR (126 MHz, $CDCl_3$/TMS) δ: 203.8, 134.8, 132.2, 129.5, 129.5, 129.2, 129.1, 128.9, 128.6, 126.1,

https://doi.org/10.1021/acs.joc.4c02516
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125.0, 124.12, 120.3, 119.3, 118.3, 116.2, 110.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{28}H_{19}N_2Br_2$ 540.9910; Found 540.9912.

*2-Methyl-1-(1,3,3-triphenylpropa-1,2-dien-1-yl)-1H-benzo[d]-imidazole (**6c**).* Following is the topical procedure for the synthesis of **6c**: **1a** (0.36 mmol), (2-methyl-1*H*-benzo[*d*]imidazole), **6b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (107.5 mg, 90%) as a yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.76 (d, J = 8.0 Hz, 1H), 7.39 (s, 9H), 7.35−7.31 (m, 3H), 7.28−7.25 (m, 2H), 7.18−7.12 (m, 4H), 2.40 (s, 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 206.2, 152.4, 142.9, 136.2, 134.8, 132.7, 129.3, 129.0, 128.9, 128.8, 125.4, 122.9, 122.5, 119.2, 119.0, 110.6, 110.1, 14.3. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{29}H_{23}N_2$ 399.1856; Found 399.1857.

*2-Phenyl-1-(1,3,3-triphenylpropa-1,2-dien-1-yl)-1H-benzo[d]-imidazole (**7c**).* Following is the topical procedure for the synthesis of **7c**: **1a** (0.36 mmol), (2-phenyl-1*H*-benzo[*d*]imidazole), **7b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (116.0 mg, 84%) as an orange oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.84 (d, J = 8.0 Hz, 1H), 7.74 (d, J = 7.5 Hz, 2H), 7.26−7.08 (m, 21H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 206.3, 153.3, 143.3, 137.3, 134.4, 133.1, 129.9, 129.7, 129.3, 129.0, 128.8, 128.7, 128.6, 128.5, 125.3, 123.5, 123.0, 120.0, 119.2, 111.2. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{34}H_{25}N_2$ 461.2012; Found 461.2012.

*1-(3,3-Diphenyl-1-(p-tolyl)propa-1,2-dien-1-yl)-1H-benzo[d]-imidazole (**8c**).* Following is the topical procedure for the synthesis of **8c**: (1,1-diphenyl-3-(*p*-tolyl)prop-2-yn-1-ol) **2a** (0.36 mmol), **1b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (77.6 mg, 65%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.83 (d, J = 8.0 Hz, 1H), 7.73 (s, 1H), 7.42−7.35 (m, 12H), 7.30−7.28 (m, 4H), 7.17−7.14 (m, 1H), 2.12 (s, 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ 200.1, 142.1, 137.4, 136.0, 132.0, 131.3, 129.9, 129.7, 129.0, 128.9, 128.7, 126.8, 124.0, 123.1, 120.6, 118.8, 112.0, 110.3, 20.1. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{29}H_{23}N_2$ 399.1856; Found 399.1848.

*1-(1-(4-Ethylphenyl)-3,3-diphenylpropa-1,2-dien-1-yl)-1H-benzo[d]imidazole (**9c**).* Following is the topical procedure for the synthesis of **9c**: (3-(4-ethylphenyl)-1,1-diphenylprop-2-yn-1-ol) **3a** (0.36 mmol), **1b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (77.6 mg, 70%) as a yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 8.04 (s, 1H), 7.87 (d, J = 8.0 Hz, 1H), 7.41−7.38 (m, 11H), 7.21−7.15 (m, 6H), 2.67 (q, J = 7.5 Hz 2H), 1.24 (t, J = 7.5 Hz 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 203.9, 145.6, 143.3, 135.3, 134.3, 131.9, 130.0, 128.9, 128.8, 128.8, 126.3, 126.2, 123.6, 122.7, 120.6, 119.2, 111.6, 111.1, 28.8, 15.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{30}H_{25}N_2$ 413.2012; Found 413.2015.

*1-(3,3-Diphenyl-1-(4-propylphenyl)propa-1,2-dien-1-yl)-1H-benzo[d]imidazole (**10c**).* Following is the topical procedure for the synthesis of **10c**: (1,1-diphenyl-3-(4-propylphenyl)prop-2-yn-1-ol) **4a** (0.36 mmol), **1b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (81.8 mg, 64%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 8.05 (s, 1H), 7.87 (d, J = 8.0 Hz, 1H), 7.45−7.37 (m, 11H), 7.22−7.16 (m, 6H), 2.60 (t, J = 7.5 Hz 2H), 1.68−1.61 (m, 2H), 0.95 (t, J = 7.0 Hz 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 203.9, 144.1, 143.3, 135.3, 129.9, 129.3, 128.9, 128.8, 128.7, 126.2, 126.0, 123.6, 122.7, 120.5, 119.2, 111.5, 111.1, 37.9, 24.5, 14.0. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{31}H_{27}N_2$ 427.2169; Found 427.2163.

*1-(1,3-Diphenyl-3-(p-tolyl)propa-1,2-dien-1-yl)-1H-benzo[d]-imidazole (**12c**).* Following is the topical procedure for the synthesis of **12c**: (1,3-diphenyl-1-(*p*-tolyl)prop-2-yn-1-ol) **5a** (0.36 mmol), **1b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (93.2 mg, 78%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 8.03 (s, 1H), 7.86 (d, J = 8.0 Hz, 1H), 7.39−7.34 (m, 8H), 7.30−7.28 (m, 4H), 7.20−7.12 (m, 5H), 2.39 (s, 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ 204.0, 143.8, 143.3, 138.9, 135.3, 134.3, 132.9, 132.1, 129.7, 129.2, 129.0, 128.9, 128.9, 128.8, 126.3, 123.7, 122.8, 120.6, 119.3, 111.6, 110.8, 21.42. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{29}H_{23}N_2$ 399.1856; Found 399.1852.

*1-(1,3-Diphenyl-3-(p-tolyl)propa-1,2-dien-1-yl)-2-methyl-1H-benzo[d]imidazole (**13c**).* Following is the topical procedure for the synthesis of **13c**: **5a** (0.36 mmol), **6b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (92.7 mg, 75%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.75 (d, J = 8.0 Hz, 1H), 7.40−7.37 (m, 5H), 7.32−7.27 (m, 5H), 7.19 (d, J = 8.5 Hz, 2H), 7.17−7.08 (m, 5H), 0.2.39 (d, J = 2.0 Hz, 6H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ 206.1, 152.5, 139.0, 136.3, 135.0, 132.9, 131.8, 129.7, 129.3, 129.0, 128.9, 128.9, 128.8, 125.4, 122.8, 122.4, 119.3, 119.0, 110.6, 109.8, 21.4, 14.3. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{30}H_{25}N_2$ 413.2012; Found 413.2010.

*1-(1,3-Diphenyl-3-(p-tolyl)propa-1,2-dien-1-yl)-2-phenyl-1H-benzo[d]imidazole (**14c**).* Following is the topical procedure for the synthesis of **14c**: **5a** (0.36 mmol), **7b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (103.4 mg, 73%) as an orange oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.89 (d, J = 8.0 Hz, 1H), 7.80 (dd, J_1 = 8.5 Hz J_2 = 1.5 Hz, 2H), 7.33−7.14 (m, 16H), 7.05 (s, 4H), 2.33 (s, 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 206.2, 153.4, 143.3, 138.7, 137.4, 134.7, 133.4, 131.6, 130.0, 129.7, 129.5, 129.3, 129.1, 128.9, 128.8, 128.7, 128.6, 128.6, 128.5, 125.3, 123.5, 123.0, 120.0, 119.2, 111.3, 111.0, 21.4. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{35}H_{27}N_2$ 475.2169; Found 475.2165.

*1-(1,3-Diphenyl-3-(p-tolyl)propa-1,2-dien-1-yl)-6-methyl-1H-benzo[d]imidazole and 1-(1,3-diphenyl-3-(p-tolyl)propa-1,2-dien-1-yl)-5-methyl-1H-benzo[d]imidazole (**15c** + **15c′**).* Following is the topical procedure for the synthesis of **15c** + **15c′**: **5a** (0.36 mmol), **3b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (86.6 mg, 70%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.74−7.72 (m, 3H), 7.57−7.53 (m, 3H), 7.51−7.49 (m, 2H), 7.40−7.39 (dd, J_1 = 2.5 Hz J_2 = 2.0 Hz,, 2H), 7.31(t, J = 7.5 Hz, 2H), 7.24 (d, J = 7.0 Hz, 1H), 7.13 (d, J = 8.5 Hz, 2H), 7.06 (d, J = 8.5 Hz, 1H) 7.03 (s, 1H) 6.35 (d, J = 4.0 Hz, 1H), 2.46 (s, 0.8H), 2.38 (s, 2.1H), 2.31 (s, 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 164.5, 164.0, 148.5, 146.3, 141.8, 141.7, 139.8, 138.6, 138.5, 137.2, 132.7, 131.8, 130.1, 130.0, 130.0, 129.5, 128.9, 128.9, 128.8, 128.4, 128.3, 128.0, 128.0, 128.0, 127.9, 127.9, 127.4, 124.1, 123.6, 122.8, 122.6, 120.6, 120.2, 111.5, 111.0, 22.0, 21.7, 21.2. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{30}H_{25}N_2$ 413.2012; Found 413.2009.

*1-(3-(4-Chlorophenyl)-1,3-diphenylpropa-1,2-dien-1-yl)-1H-benzo[d]imidazole (**16c**).* Following is the topical procedure for the synthesis of **16c**: (1-(4-chlorophenyl)-1,3-diphenylprop-2-yn-1-ol) **6a** (0.36 mmol), **1b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (69.0 mg, 55%) as a yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 8.02 (s, 1H), 7.87 (d, J = 8.5 Hz, 1H), 7.40−7.36 (m, 9H), 7.35−7.27 (m, 6H), 7.20−7.16 (m, 1H),

https://doi.org/10.1021/acs.joc.4c02516
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7.10 (d, J = 8.0 Hz, 1H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ 204.0, 143.9, 143.2, 134.9, 134.8, 134.2, 133.7, 132.5, 130.2, 129.3, 129.3, 129.2, 129.1, 128.8, 126.4, 123.8, 122.9, 120.7, 118.5, 111.5, 111.4. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{28}H_{20}N_2Cl$ 419.1310; Found 419.1303.

*5-Chloro-1-(3-(4-chlorophenyl)-1,3-diphenylpropa-1,2-dien-1-yl)-1H-benzo[d]imidazole (**17c**).* Following is the topical procedure for the synthesis of **17c**: **6a** (0.36 mmol), **4b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (67.8 mg, 50%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 8.01 (s, 1H), 7.76 (d, J = 8.5 Hz, 1H), 7.43−7.38 (m, 10H), 7.33 (d, J = 8.5 Hz, 2H), 7.28−7.27 (m, 2H), 7.26−7.25 (m, 2H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS): δ 203.9, 143.8, 135.1, 133.5, 132.2, 130.1, 129.7, 129.5, 129.5, 129.4, 129.3, 129.3, 129.2, 128.8, 126.2, 123.7, 121.5, 119.0, 111.6, 111.2. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{28}H_{19}N_2Cl_2$ 453.0920; Found 453.0921.

*1-(3-(4-Chlorophenyl)-1,3-diphenylpropa-1,2-dien-1-yl)-2-phenyl-1H-benzo[d]imidazole (**18c**).* Following is the topical procedure for the synthesis of **18c**: **6a** (0.36 mmol), **7b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (96.4 mg,65%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.89 (d, J = 8.0 Hz, 1H), 7.75 (d, J = 8.5 Hz, 2H), 7.38−7.28 (m, 9H), 7.24−7.18 (m, 7H), 7.11 (d, J = 7.5 Hz, 2H), 7.03 (d, J = 8.5 Hz, 2H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 206.3, 153.4, 143.3, 137.3, 134.6, 134.1, 133.0, 133.0, 130.1, 129.8, 129.5, 129.1, 129.0, 129.0, 128.8, 128.7, 128.7, 128.6, 126.7, 125.4, 123.7, 123.1, 120.1, 118.2, 111.7, 111.2. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{34}H_{24}N_2Cl$ 495.1623; Found 495.1623.

*2-Phenyl-1-(1-phenyl-3,3-di-p-tolylpropa-1,2-dien-1-yl)-1H-benzo[d]imidazole (**19c**).* Following is the topical procedure for the synthesis of **19c**: (3-phenyl-1,1-di-*p*-tolylprop-2-yn-1-ol) **7a** (0.36 mmol), **7b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (102.5 mg,70%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.88 (d, J = 8.0 Hz, 1H), 7.81−7.79 (m, 2H), 7.33−7.19 (m, 11H), 7.04 (s, 8H), 2.34 (s, 6H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 206.1, 153.4, 143.3, 138.6, 137.4, 133.5, 131.7, 130.1, 129.6, 129.4, 129.3, 129.2, 128.8, 128.6, 128.5, 125.3, 123.5, 123.0, 120.0, 119.1, 111.3, 110.8, 21.4. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{36}H_{29}N_2$ 489.2325; Found 489.2323.

*1-(3,3-Bis(4-methoxyphenyl)-1-phenylpropa-1,2-dien-1-yl)-1H-benzo[d]imidazole (**20c**).* Following is the topical procedure for the synthesis of **20c**: (1,1-bis(4-methoxyphenyl)-3-phenylprop-2-yn-1-ol) **8a** (0.36 mmol), **1b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (99.9 mg, 75%) as a light yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 8.05 (s, 1H), 7.87 (d, J = 8.0 Hz, 1H), 7.36−7.28 (m, 12H), 6.93 (d, J = 9.0 Hz, 4H), 3.84 (s, 6H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 203.5, 160.1, 143.8, 143.3, 134.3, 133.1, 130.2, 129.1, 128.9, 126.2, 123.6, 122.7, 120.5, 118.7, 114.3, 111.5, 110.3, 55.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{30}H_{25}N_2O_2$ 445.1911; Found 445.1910.

*1-(3,3-Bis(4-methoxyphenyl)-1-phenylpropa-1,2-dien-1-yl)-2-phenyl-1H-benzo[d]imidazole (**21c**).* Following is the topical procedure for the synthesis of **21c**: **8a** (0.36 mmol), **7b** (0.3 mmol), and $Bi(OTf)_3$ (0.045 mmol) reacted in DCE (2.0 mL) at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (126.4 mg, 81%) as a yellow oily liquid. TLC: $R_f \approx 0.6$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.89 (d, J = 8.0 Hz, 1H), 7.81 (d, J = 7.0 Hz, 2H), 7.33−7.21 (m, 11H), 7.07 (d, J = 9.0 Hz, 4H), 6.77 (d, J = 9.0 Hz, 4H), 3.80 (s, 6H) $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ 205.8, 160.0, 153.3, 143.3, 137.4, 133.6, 130.1, 130.1, 129.6, 129.3, 129.1, 128.5, 128.5, 126.8, 125.3, 123.5, 122.9, 120.0, 118.6, 114.1, 111.2, 110.5, 55.4. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{36}H_{29}N_2O_2$ 521.2224; Found 521.2221.

*(Z)-1-Styryl-1H-benzo[d]imidazole (**1d**).*[13a] According to the topical procedure for the synthesis of **1d**, **1a** (0.3 mmol), **1b** (0.3 mmol), $Pd(dppf)Cl_2$ (0.045 mmol), and Cs_2CO_3 (0.6 mmol) were added to DMF (2.0 mL) under a N_2 atmosphere and reacted at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (49.5 mg, 75%, **1a** was used; 48.2 mg, 73%, **5a** was used; 46.2 mg, 70%, **6a** was used) as a light yellow solid. Mp: 123.2−125.6 °C (Lit: Mp: 122.0 °C). TLC: $R_f \approx 0.5$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.82 (d, J = 5.5 Hz, 2H), 7.37−7.31 (m, 3H), 7.24−7.23 (m, 3H), 7.10−7.09 (m, 2H), 6.87 (d, J = 9.0 Hz, 1H), 6.59 (d, J = 9.0 Hz, 1H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS): δ 143.3, 142.1, 133.7, 133.1, 129.0, 128.6, 125.5, 123.8, 123.1, 120.5, 120.4, 110.5.

*(Z)-4-Methyl-1-styryl-1H-benzo[d]imidazole (**2d**).* According to the topical procedure for the synthesis of **2d**, **1a** (0.3 mmol), **2b** (0.3 mmol), $Pd(dppf)Cl_2$ (0.045 mmol), and Cs_2CO_3 (0.6 mmol) were added to DMF (2.0 mL) under a N_2 atmosphere and reacted at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (47.1 mg, 67%, **1a** was used; 49.2 mg, 70%, **5a** was used) as a brown solid. Mp: 124.5−126.2 °C. TLC: $R_f \approx 0.5$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.80 (s, 1H), 7.25−7.20 (m, 5H), 7.14−7.10 (m, 3H), 6.85 (d, J = 9.0 Hz, 1H), 6.56 (d, J = 9.0 Hz, 1H), 2.69 (s, 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 142.6, 141.2, 133.8, 132.8, 130.4, 129.0, 128.6, 128.5, 125.2, 123.7, 123.5, 120.6, 107.9, 16.7. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{16}H_{15}N_2$ 235.1230; Found 235.1235.

*(Z)-5-Chloro-1-styryl-1H-benzo[d]imidazole (**3d**).* According to the topical procedure for the synthesis of **3d**, **1a** (0.3 mmol), **4b** (0.3 mmol), $Pd(dppf)Cl_2$ (0.045 mmol), and Cs_2CO_3 (0.6 mmol) were added to DMF (2.0 mL) under a N_2 atmosphere and reacted at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (41.2 mg, 54%, **1a** was used; 48.2 mg, 54%, **5a** was used; 38.1 mg, 50%, **6a** was used) as a light brown oily liquid. TLC: $R_f \approx 0.5$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.82−7.78 (m, 2H) 7.27−7.22 (m, 5H), 7.07−7.03 (m, 2H), 6.84−6.79 (m, 1H), 6.62 (d, J = 9.0 Hz, 1H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 143.3, 142.9, 129.1, 129.1, 128.8, 128.8, 128.6, 126.6, 124.3, 123.7, 121.4, 120.4, 111.3, 77.4, 77.2, 76.9. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{15}H_{12}N_2$ 255.0684; Found 255.0678.

*(Z)-5,6-Difluoro-1-styryl-1H-benzo[d]imidazole (**4d**).* According to the topical procedure for the synthesis of **4d**, **1a** (0.3 mmol), (5,6-difluoro-1*H*-benzo[*d*]imidazole) **8b** (0.3 mmol), $Pd(dppf)Cl_2$ (0.045 mmol), and Cs_2CO_3 (0.6 mmol) were added to DMF (2.0 mL) under a N_2 atmosphere and reacted at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (24.6 mg, 32%) as a light yellow solid. Mp: 135.2−136.8 °C. TLC: $R_f \approx 0.5$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.75 (s, 1H), 7.38−7.31 (m, 3H), 7.24−7.23 (m, 3H), 7.10−7.09 (m, 2H), 6.87 (d, J = 9.0 Hz, 1H), 6.59 (d, J = 9.0 Hz, 1H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ 143.5 (d, J = 10.0 Hz), 133.3, 129.1, 129.0, 128.6, 127.2, 126.5, 119.8, 108.0, 107.9, 98.9, 98.7. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{15}H_{11}N_2F_2$ 257.0885; Found 257.0883.

*(Z)-6-Methyl-1-styryl-1H-benzo[d]imidazole and (Z)-5-Methyl-1-styryl-1H-benzo[d]imidazole (**5d** + **5d′**).*[12h] According to the topical procedure for the synthesis of **5d** + **5d′**, **1a** (0.3 mmol), **3b** (0.3 mmol), $Pd(dppf)Cl_2$ (0.045 mmol), and Cs_2CO_3 (0.6 mmol) were added to DMF (2.0 mL) under a N_2 atmosphere and reacted at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (53.4 mg,76%) as a light yellow liquid. TLC: $R_f \approx 0.5$. ^1H NMR (500 MHz, $CDCl_3$/TMS) δ 7.75 (d, J = 16.0 Hz, 1H), 7.69 (d, J = 8.5 Hz, 0.54H), 7.60 (s, 0.43H), 7.24−7.22 (m, 4H), 7.17−7.09 (m, 4H), 6.84 (dd, J_1 = 9.0 Hz, J_2 = 6.0 Hz, 1H), 6.55 (dd, J_1 = 9.0 Hz, J_2 = 2.6 Hz, 1H), 2.49 (d, J = 7.5 Hz,3H).$^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ 143.7, 142.1, 141.7, 141.5, 133.9, 133.9, 133.4, 132.8, 131.2, 129.0, 128.6,

https://doi.org/10.1021/acs.joc.4c02516
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128.5, 125.2, 125.1, 124.6, 120.6, 120.6, 120.3, 120.0, 110.3, 110.0, 21.9, 21.7.

*(Z)-1-(2-Methylstyryl)-1H-benzo[d]imidazole (**6d**).*[13c] According to the topical procedure for the synthesis of **6d**, (1,1-diphenyl-3-(*o*-tolyl)prop-2-yn-1-ol) **9a** (0.3 mmol), **1b** (0.3 mmol), $Pd(dppf)Cl_2$ (0.045 mmol), and Cs_2CO_3 (0.6 mmol) were added to DMF (2.0 mL) under a N_2 atmosphere and reacted at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (54.1 mg, 77%) as a yellow solid. Mp: 125.7−127.3 °C. TLC: $R_f \approx 0.5$. ^{1}H NMR (500 MHz, $CDCl_3$/TMS) δ 7.83 (s, 1H), 7.82−7.80 (m, 1H), 7.36−7.30 (m, 3H), 7.15 (t, J = 8.0 Hz 1H), 6.87 (d, J = 9.0 Hz, 1H), 6.78 (dd, J_1 = 8.5 Hz, J_2 = 2.0 Hz 1H), 6.70 (d, J = 7.5 Hz, 1H), 6.58−6.55 (m, 2H), 3.56 (s, 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 159.9, 143.3, 142.2, 135.0, 133.2, 130.0, 125.7, 123.8, 123.0, 121.2, 120.6, 120.5, 114.7, 113.6, 110.5, 55.2.

*(Z)-1-(4-Methylstyryl)-1H-benzo[d]imidazole (**7d**).*[13c] According to the topical procedure for the synthesis of **7d**, **2a** (0.3 mmol), **1b** (0.3 mmol), $Pd(dppf)Cl_2$ (0.045 mmol), and Cs_2CO_3 (0.6 mmol) were added to DMF (2.0 mL) under a N_2 atmosphere and reacted at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (55.5 mg, 79%) as a yellow semisolid. TLC: $R_f \approx 0.5$. ^{1}H NMR (500 MHz, $CDCl_3$/TMS) δ 7.83−7.81 (m, 2H), 7.37−7.35 (m, 1H), 7.32−7.30 (m, 2H), 7.03 (d, J = 8.5 Hz, 2H), 6.96 (d, J = 8.0 Hz, 2H), 6.81 (d, J = 9.0 Hz, 1H), 6.56 (d, J = 9.0 Hz, 1H), 2.29(s, 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 143.4, 142.3, 138.7, 133.2, 130.8, 129.7, 128.6, 126.0, 123.8, 123.0, 120.5, 119.8, 110.5, 21.4.

*(Z)-1-(4-Chlorostyryl)-1H-benzo[d]imidazole (**8d**).*[13a] According to the topical procedure for the synthesis of **8d**, (3-(4-chlorophenyl)-1,1-diphenylprop-2-yn-1-ol) **10a** (0.3 mmol), **1b** (0.3 mmol), $Pd(dppf)Cl_2$ (0.045 mmol), and Cs_2CO_3 (0.6 mmol) were added to DMF (2.0 mL) under a N_2 atmosphere and reacted at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (45.7 mg, 60%) as a light green semisolid. TLC: $R_f \approx 0.5$. ^{1}H NMR (500 MHz, $CDCl_3$/TMS) δ 7.83 (s, 1H) 7.77 (d, J = 6.5 Hz, 1H), 7.27 (d, J = 7.5 Hz, 3H), 7.19 (s, 1H), 7.13 (d, J = 8.5 Hz, 2H), 6.93 (d, J = 9.0 Hz, 2H), 6.84 (d, J = 9.0 Hz, 1H), 6.51 (d, J = 9.0 Hz, 1H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 134.5, 132.1, 130.0, 129.4, 129.3, 129.1, 127.6, 124.8, 124.1, 123.4, 121.0, 120.5, 110.6.

*(Z)-1-(3-Methoxystyryl)-1H-benzo[d]imidazole (**9d**).* According to the topical procedure for the synthesis of **9d**, (3-(3-methoxyphenyl)-1,1-diphenylprop-2-yn-1-ol) **11a** (0.3 mmol), **1b** (0.3 mmol), $Pd(dppf)Cl_2$ (0.045 mmol), and Cs_2CO_3 (0.6 mmol) were added to DMF (2.0 mL) under a N_2 atmosphere and reacted at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (60.8 mg, 81%) as a light yellow liquid. TLC: $R_f \approx 0.5$. ^{1}H NMR (500 MHz, $CDCl_3$/TMS) δ 7.78−7.77 (m, 1H), 7.59 (s, 1H), 7.40−7.39 (m, 1H), 7.32−7.29 (m, 2H), 7.24−7.18 (m, 2H), 7.07−6.98 (m, 3H), 6.56 (d, J = 9.0 Hz, 1H), 2.27 (s, 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 143.1, 141.9, 136.4, 133.5, 133.2, 130.7, 128.5, 128.4, 126.6, 123.8, 123.0, 121.3, 121.1, 120.5, 110.0, 20.0. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{16}H_{15}N_2O$ 251.1179; Found 251.1189.

*(Z)-4-Methyl-1-(2-methylstyryl)-1H-benzo[d]imidazole (**10d**).* According to the topical procedure for the synthesis of **10d**, **9a** (0.3 mmol), **2b** (0.3 mmol), $Pd(dppf)Cl_2$ (0.045 mmol), and Cs_2CO_3 (0.6 mmol) were added to DMF (2.0 mL) under a N_2 atmosphere and reacted at 120 °C for 24 h. Chromatographic purification of the crude product using PE/EtOAc (v:v = 4:1) afforded the desired compound (46.1 mg, 62%) as a light yellow liquid. TLC: $R_f \approx 0.5$. ^{1}H NMR (500 MHz, $CDCl_3$/TMS) δ 7.56 (s, 1H), 7.22 (d, J = 2.5 Hz, 3H), 7.20−7.18 (m, 1H), 7.05−7.04 (m, 3H), 6.97 (d, J = 9.5 Hz, 1H), 6.54 (d, J = 9.5 Hz, 1H). 2.64 (s, 3H), 2.26 (s, 3H). $^{13}C\{^1H\}$·NMR (126 MHz, $CDCl_3$/TMS) δ: 141.0, 136.3, 133.6, 131.9, 130.7, 130.5, 128.5, 128.4, 126.6, 125.7, 123.7, 123.5, 121.3, 121.0, 107.5, 29.8, 20.0. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{17}H_{17}N_2$ 249.1386; Found 249.1380.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c02516.

Detailed experimental procedures, monomer, reaction optimization, and characterization data (^{1}H and ^{13}C NMR, HRMS) for all new compounds (PDF)

Accession Codes

Deposition Numbers 2364113−2364114 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was financially supported by the Doctoral Scientific Research Startup Foundation of Liaocheng University (No. 31805403) and the Shandong Provincial Natural Science Foundation (No. ZR2023MB018).

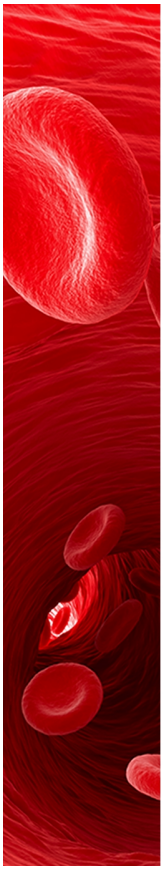
■ REFERENCES

(1) (a) Monga, J.; Ghosh, N. S.; Rani, I.; Singh, R.; Deswal, G.; Dhingra, A. K.; Grewal, A. S. Unlocking the Pharmacological Potential of Benzimidazole Derivatives: A Pathway to Drug Development. *Curr. Top. Med. Chem.* **2024**, *24*, 437−485. (b) Emami, L.; Faghih, Z.; Ataollahi, E.; Sadeghian, S.; Rezaei, Z.; Khabnadideh, S. Azole

Derivatives: Recent Advances as Potent Antibacterial and Antifungal Agents. *Curr. Med. Chem.* **2023**, *30*, 220−249.
(2) (a) Lu, T.; Lu, Z. J.; Ma, Z. X.; Zhang, Y.; Hsung, R. P. Allenamides: A Powerful and Versatile Building Block in Organic Synthesis. *Chem. Rev.* **2013**, *113*, 4862. (b) Wang, X.; Wu, Z. J.; Wang, J. α-Fluoroallenoate Synthesis via *N*-Heterocyclic Carbene-Catalyzed Fluorination Reaction of Alkynals. *Org. Lett.* **2016**, *18*, 576−579. (c) Qian, D.Y.; Wu, L.L.; Lin, Z.Y.; Sun, J.W. Organocatalyticsynthesisofchiraltetrasubstitutedallenesfromracemicpropargylicalcohols. *NatCommun* **2017**, *8*, 567. (d) Li, Y. J.; Bao, H. L. Radical transformations for allene synthesis. *Chem. Sci.* **2022**, *13*, 8491−8506. (e) Chi, Z. M.; Zhou, Y. C.; Liu, B. B.; Xu, X. J.; Liu, X. Y.; Liang, Y. M. Nickel-catalyzed regiodivergent sulfonylarylation of 1,3-enynes to access allenes and dienes. *Chem. Sci.* **2024**, *15*, 13271−13278.
(3) (a) Nagano, T. Preparation of 2-benzoylpyrrole, 2-benzoylimidazole, 2-benzoylbenzimidazole derivatives and related compounds for treatment or prevention of hyperlipidemia, arteriosclerosis, and/or metabolic syndrome: Japan. JP2006182668 [A], 2006-07−13. (b) Ushiroda, K.; Maruta, K.; Takazawa, T.; Nagano, T.; Taiji, M.; Kohno, T.; Sato, Y.; Horai, S.; Yanagi, K.; Nagata, R. Synthesis and pharmacological evaluation of novel benzoylazole-based PPAR α/γ activators. *Bioorg. Med. Chem. Lett.* **2011**, *21*, 1978−1982. (c) Nagata, R.; Ushiroda, K.; Maruta, K.; Watanabe, K. Preparation of benzoylpyrrole derivatives as PPAR agonist: World Intellectual Property Organization. WO2005012245 [A1], 2005−02−10.
(4) (a) Jones, B. C. N. M.; Silverton, J. V.; Simons, C.; Megati, S.; Nishimura, H.; Maeda, Y.; Mitsuya, H.; Zemlicka, J. Synthesis, absolute configuration, and enantioselectivity of antiretroviral effect of (R)-(−)- and (S)-(+) cytallene. Lipase-catalyzed enantioselective acylations of (+-)-N_4-acylcytallenes. *J. Med. Chem.* **1995**, *38*, 1397−1405. (b) Zhu, Y. L.; Pai, S. B.; Liu, S. H.; Grove, K. L.; Jones, B. C. N. M.; Simons, C.; Zemlicka, J.; Cheng, Y. C. Inhibition of replication of hepatitis B virus by cytallene in vitro. *Antimicrob. Agents Chemother.* **1997**, *41*, 1755−1760.
(5) (a) Linden, P. A.; Zimmermann, J.; Brixner, T.; Holt, N. E.; Vaswani, H. M.; Hiller, R. G.; Fleming, G. R. Transient Absorption Study of Peridinin and Peridinin-Chlorophyll a-Protein after Two-Photon Excitation. *J. Phys. Chem. B* **2004**, *108*, 10340−10345. (b) Miller, D. J.; Catmull, J.; Puskeiler, R.; Tweedale, H.; Sharples, F. P.; Hiller, R. G. Reconstitution of the Peridinin-chlorophyll a Protein (PCP): Evidence for Functional Flexibility in Chlorophyll Binding. *Photosynth. Res.* **2005**, *86*, 229−240.
(6) Li, Y.; Chen, J.; Qiu, R.; Wang, X.; et al. Cesium hydroxide-catalyzed isomerization of terminal alkynes for the synthesis of *O*-allenes and *N*-allenes. *Tetrahedron Lett.* **2015**, *56*, 5504−5507.
(7) (a) Wang, X.; Wang, X. S.; Shu, S. B.; Liu, J. C.; Liu, Q.; Wang, T.; Zhang, Z. M. N^1-Allylation of Arylhydrazines via a Palladium-Catalyzed Allylic Substitution. *Org. Lett.* **2023**, *25*, 4880−4885. (b) Kocsis, M.; Otvös, S. B.; Samu, G. F.; Fogarassy, Z. P.; Kukovecz, A. B.; Kónya, Z.; Sipos, P.; Pálinkó, I.; Varga, G. Copper-Loaded Layered Bismuth Subcarbonate-Efficient Multifunctional Heterogeneous Catalyst for Concerted C−S/C−N Heterocyclization. *ACS Appl. Mater. Interfaces* **2021**, *13*, 42650−42661. (c) Laha, J. K.; Hunjan, M. K.; Bhimpuria, R. A.; Kathuria, D.; Bharatam, P. V. Geometry Driven Intramolecular Oxidative Cyclization of Enamides: An Umpolung Annulation of Primary Benzamides with Acrylates for the Synthesis of 3-Methyleneisoindolin-1-ones. *J. Org. Chem.* **2017**, *82*, 7346−7352. (d) Nikolić, A. M.; Ajdačić, V.; Opsenica, I. M. Palladium-catalyzed N-Arylation of 1-substituted-1H-tetrazol-5-amines. *J. Organomet. Chem.* **2019**, *880*, 134−142. (e) Li, Y. Y.; Gao, B. Rhodium-Catalyzed Enantioselective N-Allylation of Sulfoximines. *Org. Lett.* **2023**, *25*, 2756−2760. (f) Liu, Y.; Shing, K. P.; Lo, V. K. Y.; Che, C. M. Iron- and Ruthenium-Catalyzed C-N Bond Formation Reactions. Reactive Metal Imido/Nitrene Intermediates. *ACS. Catal.* **2023**, *13*, 1103−1124.
(8) (a) Li, Y. A.; Wu, G.; Li, J. Palladium-Catalyzed *N*-Alkenylation of *N*-Aryl Phosphoramidates with Alkenes. *Molecules.* **2023**, *28*, 4466. (b) Karunakar, G. V.; Raju, C. E.; Himaja, K. S. B. N.; Kumar, P. B.; Sridhar, B. Silver-catalyzed Regioselective Synthesis of *N*-aryl-1H-pyrazolyl-substituted Benzenesulfonamides from Ynamides and Pyrazoles. *Chem. Asian J.* **2023**, *18*, No. e202300346. (c) Athira, C.; Sreenithya, A.; Hadad, C. M.; Sunoj, R. B. Cooperative Asymmetric Dual Catalysis Involving a Chiral *N*-Heterocyclic Carbene Organocatalyst and Palladium in an Annulation Reaction: Mechanism and Origin of Stereoselectivity. *ACS Catal.* **2023**, *3*, 1133−1148.
(9) (a) Wu, G.; Su, W. P. Regio- and Stereoselective Direct *N*-Alkenylation of Indoles via Pd-Catalyzed Aerobic Oxidation. *Org. Lett.* **2013**, *15*, 5278−5281. (b) Pankajakshan, S.; Ang, W. L.; Sreejith, S.; Stuparu, M. C.; Loh, T. P. Aerobic Copper Catalysis for Tandem Oxy-N-alkenylation of [1,2,3]Triazolo[1,5-a]pyridines. *Adv. Synth. Catal.* **2016**, *358*, 3034−3038.
(10) Kikuchi, J.; Nakajima, R.; Yoshikai, N. Three-component *N*-alkenylation of azoles with alkynes and iodine(III) electrophile: synthesis of multisubstituted *N*-vinylazoles. *Beilstein J. Org. Chem.* **2024**, *20*, 891−897.
(11) Alshreimi, A. S.; Zhang, G. Q.; Shim, E. J.; Wink, D. J.; Anderson, L. L. Gold-Catalyzed of and the Use of Alkenyl Gold Intermediates in the Synthesis of 2-Amino-1-pyrrolines. *ACS Catal.* **2024**, *14*, 2229−2234.
(12) (a) Lai, T. T.; Xie, D.; Zhou, C. H.; Cai, G. X. Copper-Catalyzed Inter/Intramolecular *N*-Alkenylation of Benzimidazoles via Tandem Processes Involving Selectively Mild Iodination of sp^3 C-H Bond at α-Position of Ester. *J. Org. Chem.* **2016**, *81*, 8806−8815. (b) Zhou, M. W.; Tan, X. F.; Hu, Y. M.; Shen, H. C.; Qian, X. H. Highly Chemo- and Regioselective Vinylation of *N*-Heteroarenes with Vinylsulfonium Salts. *J. Org. Chem.* **2018**, *83*, 8627−8635. (c) Wang, Z. M.; Bao, W. L.; Jiang, Y. *L*-Proline promoted Ullmann-type reaction of vinyl bromides with imidazoles in ionic liquids. *Chem. Commun.* **2005**, *22*, 2849−2851. (d) Bao, W. L.; Liu, Y. Y.; Lv, X. Mild Copper(I) Iodide/β-Keto Ester Catalyzed Coupling Reactions of Styryl Bromides with Phenols, Thiophenols, and Imidazoles. *Synthesis* **2008**, *12*, 1911−1917. (e) Lam, P. Y. S.; Vincent, G.; Clark, C. G.; Deudon, S.; Jadhav, P. K. Copper-catalyzed general C-N and C-O bond cross-coupling with arylboronic acid. *Tetrahedron. Lett.* **2001**, *42*, 3415−3418. (f) Bourgeois, P.; Lucrece, J.; Dunoguès, J. Nitrogen heterocyclic addition to arylalkynes: synthesis of Nstyrylazoles. *J. Heterocyclic. Chem.* **1978**, *15*, 1543−1545. (g) Tzalis, D.; Koradin, C.; Knochel, P. Cesium hydroxide catalyzed addition of alcohols and amine derivatives to alkynes and styrene. *Tetrahedron Lett.* **1999**, *40*, 6193−6195. (h) Nishino, S.; Sudo, K.; Kurahashi, T. Nickel-Photoredox-Catalyzed Stereoconvergent Coupling of Alkenyl Halides and Nitrogen-Containing Heterocycles. *Org. Lett.* **2024**, *26*, 4049−4054.
(13) (a) Mao, J. C.; Xie, G. L.; Zhan, J. M.; Hua, Q. Q.; Shi, D. Q. Iron-Catalyzed Cross-Coupling Reaction of Vinyl Bromides or Chlorides with Imidazoles in the Absence of Ligands and Additives. *Adv. Synth. Catal.* **2009**, *351*, 1268−1272. (b) Sun, M. M.; Wu, H. D.; Zheng, J. N.; Bao, W. L. Palladium-Catalyzed Oxidative Intramolecular C-C Bond Formation via Double sp^2 C-H Activation between the 2-Position of Imidazoles and a Benzene Ring. *Adv. Synth. Catal.* **2012**, *354*, 835−838. (c) Reddy, V. P.; Iwasaki, T.; Kambe, N. Synthesis of imidazo and benzimidazo[2,1-a]isoquinolines by rhodium-catalyzed intramolecular double C-H bond activation. *Org. Biomol. Chem.* **2013**, *11*, 2249−2253. (d) Lu, L. H.; Yan, H.; Liu, D. F.; Rong, G. W.; Mao, J. C. Transition-Metal-Free Synthesis of *N*-(1-Alkenyl) imidazoles by Potassium Phosphate-Promoted Addition Reaction of Alkynes to Imidazoles. *Chem.—Asian J.* **2014**, *9*, 75−78. (e) Pan, X. X.; Wan, X. L.; Yu, X.; Zhang, H.; Xie, W. Q. Decarboxylative Hydroamination of 3-Arylpropiolic Acids with *N*-Heterocycles under Transition-Metal-Free Conditions. *Synthesis* **2014**, *46*, 2057−2064.
(14) (a) Liu, Z. Q.; Liang, C.; Luo, Z.; Wu, Y. F.; Hong, C. M.; Li, Q. H.; Liu, T. L. Transfer Vinylation and Dienylation via Rhodium(I)-Catalyzed Deketonation of Allylic Alcohols. *ACS Catal.* **2022**, *12*, 7030−7036. (b) Yao, R. M.; Jing, Y. R.; Liu, J. H.; Liu, Y.; Wang, T. J.; Ke, Z. F. A substrate-dependent mechanism for nickel-catalyzed *N*-allylation with allylic alcohols: nucleophilic attack vs. reductive elimination. *Org. Chem. Front.* **2023**, *10*, 2721−2727.

https://doi.org/10.1021/acs.joc.4c02516
J. Org. Chem. 2025, 90, 2602−2612

(c) Ning, C.; Yu, Z. Q.; Shi, M.; Wei, Y. Palladium-catalyzed selective C-C bond cleavage of keto-vinylidenecyclopropanes: construction of structurally rich dihydrofurans and tetrahydrofurans. *Chem. Sci.* **2024**, *15*, 9192−9200.
(15) Hong, S. Y.; Hwang, Y.; Lee, M.; Chang, S. Mechanism-Guided Development of Transition-Metal-Catalyzed C-N Bond-Forming Reactions Using Dioxazolones as the Versatile Amidating Source. *Acc. Chem. Res.* **2021**, *54*, 2683−2700.
(16) Shen, G. D.; Wang, Z. Y.; Huang, X. Q.; Hong, M.; Fan, S. H.; Lv, X. Palladium/Copper Cocatalyzed C-H Activation and C-C Bond Regioselective Cleavage Reaction for the Synthesis of Fused Chromeno Quinolines. *Org. Lett.* **2020**, *22*, 8860−8865.
(17) Song, F.; Wang, B. Q.; Shi, Z. J. Transition-Metal-Catalyzed C-C Bond Formation from C-C Activation. *Acc. Chem. Res.* **2023**, *56*, 2867−2886.
(18) (a) Kumar, G. R.; Rajesh, M.; Lin, S.; Liu, S. Propargylic Alcohols as Coupling Partners in Transition-Metal-Catalyzed Arene C-H Activation. *Adv. Synth. Catal.* **2020**, *362*, 5238−5256. (b) Lutz, M. D. R.; Morand, B. Metal-Catalyzed Carbon-Carbon Bond Cleavage of Unstrained Alcohols. *Chem. Rev.* **2021**, *121*, 300−326. (c) Huang, K.; Sheng, G. R.; Lu, P.; Wang, Y. G. From 1-Sulfonyl-4-aryl-1,2,3-triazoles to 1-Allenyl-5-aryl-1,2,3-triazoles. *J. Org. Chem.* **2017**, *82*, 5294−5300.
(19) (a) Han, Y. P.; Song, X. R.; Qiu, Y. F.; Zhang, H. R.; Li, L. H.; Jin, D. P.; Sun, X. Q.; Liu, X. Y.; Liang, Y. M. Lewis Acid Catalyzed [4 + 3] Cycloaddition of Propargylic Alcohols with Azides. *Org. Lett.* **2016**, *18*, 940−943. (b) Jana, S.; Dey, A.; Singsardar, M.; Bagdi, A. K.; Hajra, A. $Zn(OTf)_2$-Catalyzed Synthesis of Imidazole-Substituted Allenes. *J. Org. Chem.* **2016**, *81*, 9489−9493.
(20) Cabrero-Antonino, J. R.; Leyva-Pérez, A.; Corma, A. Beyond Acid Strength in Zeolites: Soft Framework Counteranions for Stabilization of Carbocations on Zeolites and Its Implication in Organic Synthesis. *Angew. Chem., Int. Ed.* **2015**, *54*, 5658−5661.


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https://doi.org/10.1021/acs.joc.4c02516
J. Org. Chem. 2025, 90, 2602−2612